

## 4 Data Formats: XML and JSON

</> Code



### Learning Objectives

- Distinguish between XML and JSON as structured data formats and identify when each is typically encountered
- Parse XML documents using BeautifulSoup's tree navigation methods
- Navigate nested JSON structures using Python's native lists and dictionaries
- Convert nested data structures into pandas DataFrames for analysis
- Handle common edge cases in structured data parsing



### Companion Notebook

Run this chapter's code as you read: open the [companion notebook](#) (see [Appendix A](#) for all of them).

### Table of contents

[4.1 Why Data Formats Matter](#)

[4.2 XML and BeautifulSoup](#)

[4.3 JSON and Python Data Structures](#)

[4.4 When to Use Which](#)

[4.5 Recommended Exercises](#)

[4.6 Additional Exercises](#)

[4.7 Social History and Public Interest](#)

[4.8 Common Issues to Debug](#)

[4.9 Key Takeaways](#)

[4.10 Further Reading](#)



Edit this page

1

[Report an issue](#)